Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/07/2018
PI Name:	Ribaric, Nick M.E.E.		
Project Title:	Multispectral Imaging for the Early Detection of Ophthalmological Injury in Idiopathic Intracranial Hypertension		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	nickzr@annidis.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	
Organization Name:	Annidis Inc.		
PI Address 1:	2885 Sanford Avenue SW		
PI Address 2:	#29867		
PI Web Page:			
City:	Grandville	**State:**	MI
Zip Code:	49418	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI-RFA-SMARTCAP
Start Date:	07/01/2015	**End Date:**	03/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2017 (previously 12/31/2016) per NSBRI (Ed., 12/27/16) NOTE: End date changed to 12/31/2016 per NSBRI (Ed., 6/16/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bershad, Eric M.D. (Baylor College of Medicine)		
Grant/Contract No.:	NCC 9-58-SMST00009		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The main purpose of our study is to evaluate the clinical usefulness of multispectral imaging (MSI) to detect early ophthalmological abnormalities in patients with idiopathic intracranial hypertension (IIH). We will compare the findings obtained by MSI to standard objective clinical methods used for evaluation of IIH including optical coherence tomography (OCT), fundus photography, and Frisen (papilledema) grading. Outcome measures will include qualitative and quantitative assessment of choroidal thickness, choroidal folds, Frisen grade, and retinal fiber layer hemorrhages. We will develop a novel scoring system to quantify the data obtained from MSI to allow for a clinically-useful measure of retinal, choroidal, and optic nerve head abnormalities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our data represents the largest series of patients to date with chronically elevated intracranial pressure (ICP) with MSI technology. This may provide a novel method to diagnose and follow the efficacy of treatment, and understand how changes in MSI parameters correlate with other clinically indicated testing including optical coherence tomography, fundus photography, optic nerve sheath ultrasound, lumbar puncture, and visual fields. This test may prove to be a highly useful complementary diagnostic modality for patients with conditions that include chronically elevated ICP. The database will further be evaluated to start development of methods for quantitative assessment of pathologies associated with elevated ICP.
Task Progress:	As previously discussed, enrollment was successfully completed in IIH patients and healthy controls. The database construction is nearly completed, and data analysis is ongoing. A demonstration of technology with NASA flight physicians and key Human Research Program (HRP) personnel was completed in May 2017.
Bibliography Type:	Description: (Last Updated:)